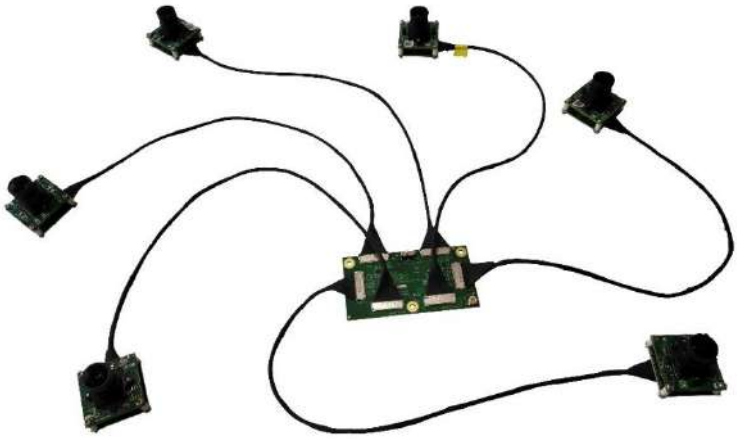
e-CAM80_CUOAGX

eCAM_Argus_Camera Application User Manual



e-con Systems

Think Camera. Think e-con.

Version 1.3

e-con Systems

6/18/2024

Disclaimer

e-con Systems reserves the right to edit/modify this document without any prior intimation of whatsoever.

© Copyright e-con Systems. 2024. All rights reserved.

Contents

INTRODUCTION TO E-CAM80_CUOAGX 3

DESCRIPTION 3

LAUNCHING THE APPLICATION 5

APPLICATION FEATURES 7

EXPOSURE TIME RANGE 9
GAIN RANGE 9
ISP DIGITAL GAIN RANGE 10
SENSOR MODE INDEX 10
FRAME RATE RANGE 11
DE-NOISE MODE 11
DE-NOISE STRENGTH 12
EDGE ENHANCE MODE 12
EDGE ENHANCE STRENGTH 12
AE ANTIBANDING MODE 13
AWB MODE 13
OUTPUT SIZE 14
AE LOCK 14
AWB LOCK 15
STILL FILE TYPE 15
VIDEO RECORDING 15
MULTI SESSION 16
MULTICAMERA SYNCHRONIZATION 16

COMPARING ECAM_ARGUS_CAMERA WITH ARGUS_CAMERA BY NVIDIA 18

TROUBLESHOOTING 19

FAQ 20

WHAT'S NEXT? 21

GLOSSARY 22

SUPPORT 23

Introduction to e-CAM80_CUOAGX

e-CAM80_CUOAGX is a new MIPI camera board which helps to connect upto six IMX415 camera modules in 2-lane configuration and upto 4 IMX415 camera modules in 4-lane configuration to the Jetson AGX Orin™ development kit. e-con Systems, a company with over two decades of experience in designing, developing, and manufacturing OEM cameras. The prebuilt driver for this camera along with the camera board is provided by e-con Systems.

e-CAM80_CUOAGX is an 8 MP color camera with S-mount (also known as M12 board lens) lens holder. The S-mount is small form-factor lens mounts for board cameras. e-CAM80_CUOAGX camera contains 1/2.8" IMX415 sensor from STARVIS family of CMOS image sensor from Sony ®. All the six cameras were interfaced to the J509 connector of the Jetson AGX Orin™ development kit using the e-CAM30_HEXCUXVR_BASE_BRD base board.

This document describes the usage of eCAM_Argus_Camera application, and the special features of eCAM_Argus_Camera application when it is used with e-CAM80_CUOAGX.

Description

Each of the IMX415 based camera in e-CAM80_CUOAGX can stream in 4-lane or 2-lane mode in NVIDIA® Jetson AGX Orin™ Development Kit.

The supported resolutions and frame rate range in 4-lane mode and 2-lane mode are listed in below tables.

Table 1: Supported Resolutions and Frame rate range in 4-Lane Mode

S.No	Resolution	Bits per pixel (bpp)	Frame rate range (fps)
1	3840x2160	12	30
2	3840x2160	10	60
3	1920x1080*	12	60
4	1280x720**	12	89

Table 2: Supported Resolutions and Frame rate range in 2-Lane Mode

S.No	Resolution	Bits per pixel (bpp)	Frame rate range (fps)
1	3840x2160	12	16
2	3840x2160	10	30
3	1920x1080*	12	32
4	1280x720**	12	48

(*) - 1944 x 1096 is cropped to 1920 x 1080 on the application end to skip the dummy bytes. 1944 x 1096 will be referred to as 1920 x 1080 throughout the document.

© Copyright e-con Systems. 2024. All rights reserved.

(**) - 1296 x 732 is cropped to 1280 x 720 on the application end to skip the dummy bytes. 1296 x 732 is obtained from the sensor through centre cropping and binning. 1296 x 732 will be referred to as 1280 x 720 throughout the document.

The camera controls of e-CAM80_CUOAGX are as follows:

- Exposure Time Range
- Gain Range
- ISP Digital Gain Range
- Sensor Mode Index
- Frame rate range
- De-Noise Mode
- De-Noise Strength
- Edge Enhance Mode
- Edge Enhance Strength
- AE antibanding Mode
- AWB Mode
- AE Lock
- AWB Lock
- Output Size

The eCAM_Argus_Camera is a sample libArgus based camera application used to demonstrate the functioning of e-CAM83_CUMI415 module boards in Jetson AGX Orin™.

Using eCAM_Argus_Camera application, you can perform the following:

- Enumerate and list all video capture devices connected.
- Stream all available resolutions if different resolutions are supported by the device.
- Change controls for all available cameras.
- Capture images in RAW and JPEG formats.
- Record H264, H265 encoded videos on Jetson AGX Orin™ board.

e-con Systems provides prebuilt binary of the application along with source code. Please refer to the *e-CAM80_CUOAGX_Release_Notes_<REV>.pdf* for the compatible Linux distribution version (L4T version).

Launching the Application

This section describes how to launch the eCAM_Argus_Camera application.

The commands and output messages in this manual are represented by different colors as listed in below table.

Table 3: Notation of Colors

Color	Notation
Blue	Commands running in development board
Red	Output message in development board

The steps to launch the application are as follows:

1. Connect the e-CAM80_CUOAGX to the Jetson AGX Orin™ development kit.
2. Run the following command to check whether all the cameras are initialized.

```
$ sudo dmesg | grep "Detected eimx415 sensor"
```

The output message appears as shown below.

```
Detected eimx415 sensor
```

The output messages indicate that all the cameras are initialized properly.

3. Run the following command to check the presence of video node.

```
$ ls /dev/video*
```

The output message for camera setup appears as shown below.

```
/dev/video0
```

Note: The number of video node entries will be the same as the number of connected cameras.

4. Run the following commands to achieve maximum frame rate range before launching the eCAM_argus_camera application in the Jetson™ board.

```
$ sudo jetson_clocks
$ sudo nvpmodel -m 0
```

5. Run the following command to launch the eCAM_argus_camera application.

```
$ eCAM_argus_camera --device=<n>
```

Where **<n>** refers to the device node.

When entering the command in terminal, the screen appears as shown below.

© Copyright e-con Systems. 2024. All rights reserved.

Figure 1: Terminal Window

Once the application is launched, the control window and video display window appear as shown below.

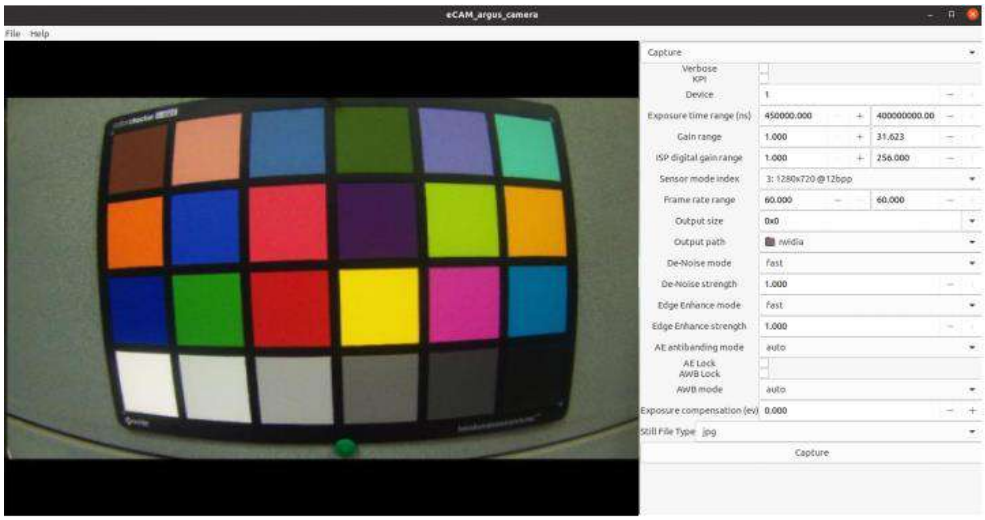


Figure 2: Initial Control Window and Video Display Window Appearance

Application Features

This section describes the controls that are supported in the current version of eCAM_Argus_Camera application for e-CAM80_CUOAGX.

The sensor features that are supported in the current version of eCAM_Argus_Camera application are as follows:

- Exposure Time Range
- Gain Range
- Sensor Mode Index
- Frame rate range

The camera ISP features supported in the current release version of eCAM_Argus_Camera application are as follows:

- ISP Digital Gain Range
- De-Noise Mode
- De-Noise Strength
- Edge Enhance Mode
- Edge Enhance Strength
- AE Antibanding Mode
- AWB Mode
- Output Size
- AWB Lock
- AE Lock
- Still File Type
- Video Recording
- Multi Session

In eCAM_Argus_Camera application, the sensor and camera ISP features appear as shown below.

© Copyright e-con Systems. 2024. All rights reserved.

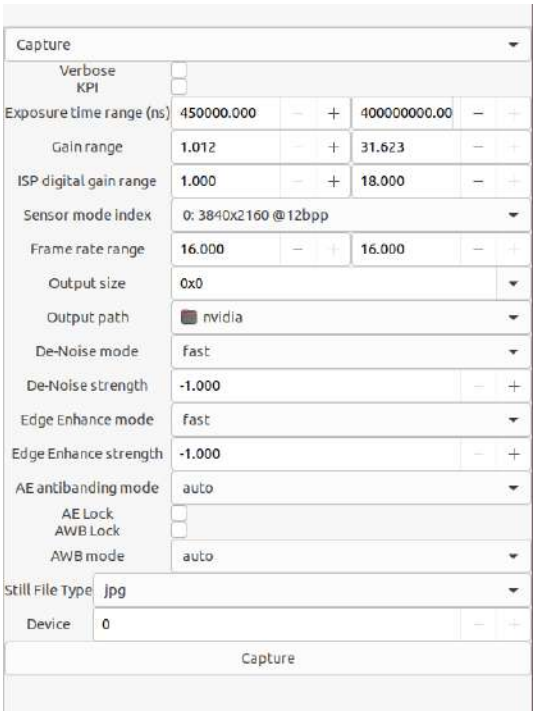


Figure 3: Controls of e-CAM80_CUOAGX

You can click the respective control box and configure the preview settings according to your needs. The value being set will be displayed in the respective text box. Once the value is set to the respective control box, you can notice the change in preview's property.

The values of e-CAM80_CUOAGX controls are listed in below table.

Table 4: Values of e-CAM80_CUOAGX Controls

Controls	Minimum Value	Maximum Value	Default Value	Manual Control	Auto Control
Exposure time range (ns)	450000	400000000	Auto	Yes	Yes
Gain range (A_f)	1.012	31.623	-	Yes	Yes
ISP digital gain range	1	18	-	Yes	Yes
Sensor mode index	0	3	0	Yes	No
Frame rate range (fps)	2.5	based on sensor mode	based on sensor mode	Yes	Yes
De-Noise mode	Off	High Quality	Fast	Yes	No
De-Noise strength	-1.00	1.00	-1.00	Yes	No
Edge enhance mode	Off	High Quality	Fast	Yes	No
Edge enhance strength	-1.00	1.00	-1.00	Yes	No
AE antibanding mode	Off	60 Hz	Auto	Yes	Yes
AWB mode	Off	Manual	Auto	Yes	Yes
AWB Lock	-	-	Not selected	-	-
AE Lock	-	-	Not selected	-	-

Note: The application currently does not support focus position control since the camera has a custom lens which uses only manual focus.

Exposure Time Range

The exposure time range can be changed from a minimum value of 450000 to a maximum value of 400000000 by manually inserting the values or selecting the value in **Exposure time range (ns)** box. The exposure time is represented in Nanoseconds (ns) scale. The default exposure time range set by the application depends upon the luminance of the source. However, to set a specific exposure time range value, you can manually set both the lower and higher range fields to that same specific value in **Exposure time range (ns)** box. You can view the screen similar to the screen shown below.

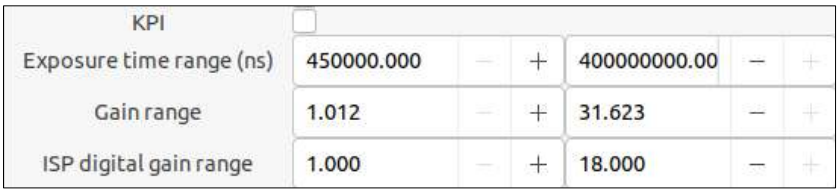


Figure 4: Exposure Time Range

Note: The exposure control is limited to the frame rate range. For example, the maximum exposure that can be achieved is limited to maximum of 40 ms even a larger exposure range value is specified in the **Exposure time range (ns)** box, when the frame rate range control is set to 25 fps as minimum value.

Gain Range

The gain range which corresponds to sensor gain range can be changed from a minimum value of 1.012 to a maximum value of 31.623 is represented in amplification factor (A_f). IMX415 can support a sensor gain range of 1 to 72 dB corresponding to 0 to 3981.0717 (A_f). However, in e-CAM80_CUOAGX, sensor gain range has been restricted to sensor analog gain, that is. 0 - 30 dB, including the image quality and other performance metrics. The default gain value set depends upon the luminance of the source. However, to set a specific constant sensor gain value, you can manually set both the lower and higher range fields to that same specific value in **Gain range** box. You can view the screen similar to the screen shown below.

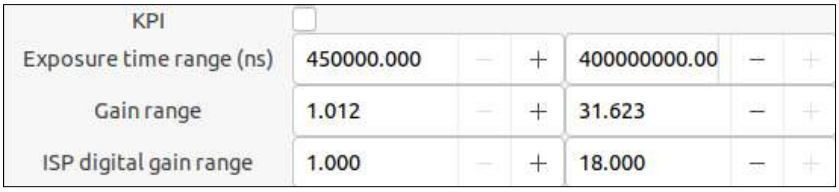


Figure 5: Gain Range in Jetson AGX Orin™

© Copyright e-con Systems. 2024. All rights reserved.

Note: The sensor gain range and exposure time range changes are local. On changing the modes, the corresponding values will be discarded, and the default range values will be applied.

ISP Digital Gain Range

The ISP gain control can be changed from a minimum value of 1 to a maximum of 18. This gain corresponds to the digital gain additionally provided by the ISP (over and above the sensor gain) and the default value depends upon the luminance of the source. Jetson AGX Orin™ ISP can support a digital gain range of 1 to 256 (A_f) corresponding to 0 to 48.16 dB. However, in e-CAM80_CUOAGX, ISP gain has been restricted to 1 - 18 (A_f), to increase the image quality and other performance metrics. To manually set the ISP digital gain, you can change the lower range value and the higher range value to a constant in the **ISP digital gain range** box. You can view the screen similar to the screen shown below.

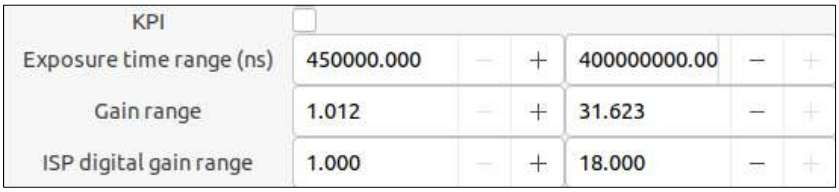


Figure 6: ISP Digital Gain Range

Sensor Mode Index

The sensor mode index contains all the combinations of resolutions, frame rate and bpp (bits per pixel) that the camera sensor outputs.

The sensor modes currently supported by the e-CAM80_CUOAGX are listed in below table.

Table 5: Supported Resolutions and Frame rate range in 4-Lane Mode

Sensor mode	Resolution	Bits per pixel (bpp)	Frame rate range (fps)
0	3840x2160	12	30
1	3840x2160	10	60
2	1920x1080*	12	60
3	1280x720**	12	89

Table 6: Supported Resolutions and Frame rate range in 2-Lane Mode

Sensor mode	Resolution	Bits per pixel (bpp)	Frame rate range (fps)
0	3840x2160	12	16
1	3840x2160	10	30
2	1920x1080*	12	32

3	1280x720**	12	48

Note: The output from the NVIDIA® ISP is NV12 format.

You can change the sensor modes from the **Sensor mode index** drop-down list box. When you click the **Sensor mode index** drop-down list box, you can view the list of sensor modes as shown below.

Figure 7: Sensor Mode Index

Frame rate range

The frame rate range control can be changed from a minimum value to the maximum value specified by the sensor mode (Refer *Table 5 and Table 6*). The frame rate range can be changed by manually inserting the value or selecting the value in **Frame rate range** box. Here we can be able to set the frame rate range so that based on the exposure time change, the frame rate range is changed in that particular frame rate range. You can view the screen similar to the screen shown below.

Figure 8: Frame rate range

Note: Frame rate range displayed (when KPI is enabled) is the number of frames from all streaming cameras per second. For example, 3840 x 2160 at 30 fps mode, when two cameras are streaming, the FPS displayed will be (30*2) = 60 fps. Also, the FPS displayed will not go beyond the refresh rate of the monitor used.

De-Noise Mode

The eCAM_argus_camera application supports the following de-noise modes:

- Off
- Fast
- High Quality

You can select the required de-noise mode from the **De-Noise mode** drop-down list box. The default de-noise mode is fast. When you click the **De-Noise mode** drop-down list box, the supported de-noise modes will be listed as shown below.

© Copyright e-con Systems. 2024. All rights reserved.

Figure 9: De-Noise Mode

De-Noise Strength

You can change the de-noise strength value from a minimum value of -1.0 to a maximum value of 1.0. The default de-noise strength value is 1.0 and can be changed by manually inserting the values or selecting the values in **De-Noise strength** box. You can view the screen similar to the screen shown below.

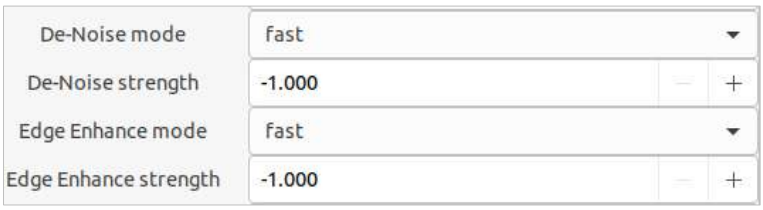


Figure 10: De-Noise Strength

Edge Enhance Mode

The eCAM_argus_camera application supports the following edge enhance modes:

- Off
- Fast
- High Quality

You can select the required edge enhancement mode from the **Edge Enhance mode** drop-down list box. The default edge-enhance mode is fast. When you click the **Edge Enhance mode** drop-down list box, the supported edge enhancement modes will be listed as shown below.

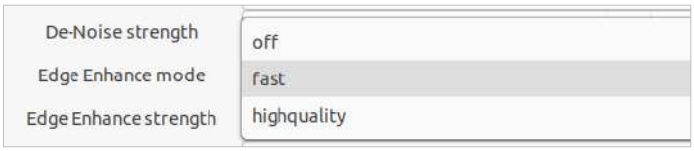


Figure 11: Edge Enhance Mode

Edge Enhance Strength

You can change the edge enhance strength value from a minimum value of -1.0 to a maximum value of 1.0. The default edge enhance strength value is 1.0 and can be changed by manually inserting the values or selecting the values in **Edge Enhance strength** box. You can view the screen similar to the screen shown below.

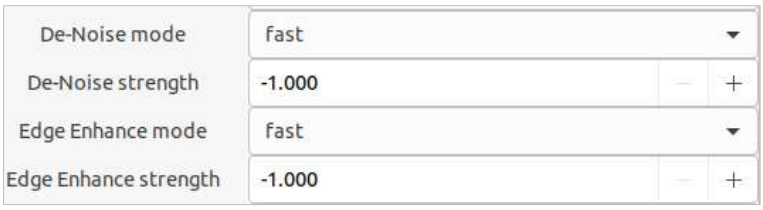


Figure 12: Edge Enhance Strength

AE Antibanding Mode

The eCAM_Argus_Camera application supports the AE antibanding modes as follows:

- Off
- Auto
- 50 Hz
- 60 Hz

By default, the AE antibanding mode is auto. You can select the required AE antibanding mode from the **AE antibanding mode** drop-down list box. When you click the **AE antibanding mode** drop-down list box, the list of AE antibanding modes appear as shown below.

Figure 13: AE Antibanding Mode

AWB Mode

The eCAM_Argus_Camera application automatically detects the scene temperature and adjusts the white balance. The application supports the following AWB modes:

- Off
- Auto
- Incandescent
- Fluorescent
- Warm Fluorescent
- Daylight
- Cloudy Daylight
- Twilight
- Shade

© Copyright e-con Systems. 2024. All rights reserved.

By default, auto mode is enabled. You can select the required AWB modes from the **AWB mode** drop-down list box. When you click the **AWB mode** drop-down list box, the list of AWB modes appear as shown below.

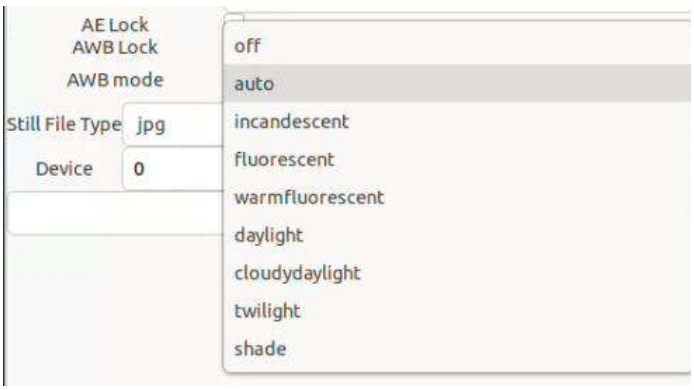


Figure 14: AWB Mode

Output Size

When you click the **Output size** drop-down list box, you can view the supported output sizes as shown below.

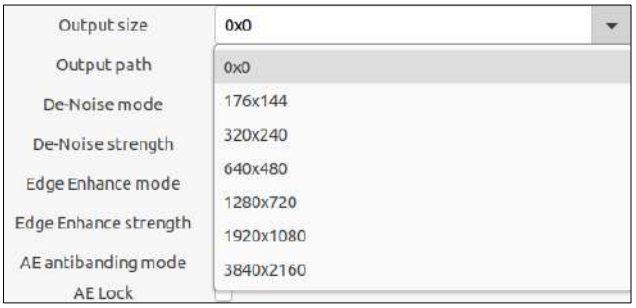


Figure 15: Output size

Note: 0x0 Output size denotes the original size

AE Lock

The AE lock control is enabled during the capture and video module of eCAM_Argus_Camera. When you select the **AE Lock** check box, it locks the exposure to current values. By default, the AE lock is set when the application runs in multi session mode. The AE Lock will appear as shown below.

Figure 16: AE Lock

AWB Lock

When you select the **AWB Lock** check box, it locks the AWB to current value. The AWB Lock option will appear as shown below.

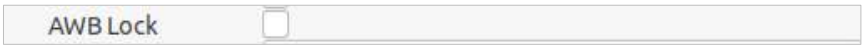


Figure 17: AWB Lock

Still File Type

There are two types of still capture supported in eCAM_Argus_Camera. You can select the type of still image capture to be JPEG compressed or uncompressed ISP processed YUV frame using the **Still File Type** drop-down list box provided in the Capture module of eCAM_Argus_Camera.

When you click the **Still File Type** drop-down list box, you can view the supported still file type as shown below.

Figure 18: Still File Type

Video Recording

The eCAM_Argus_Camera also provides the feature for recording video.

When you click the **Video Format** drop-down list box, the list of supported formats for Jetson AGX Orin™ will appear as shown in below figures.

Figure 19: Video Recording

The supported video formats supported by eCAM_Argus_Camera are as follows:

- H264
- H265

When you click the **Video File Type** drop-down list box, the list of supported file types will appear as shown below.

© Copyright e-con Systems. 2024. All rights reserved.

Figure 20: Video File Type

The supported video file types are as follows:

- MP4
- 3GP
- AVI
- MKV

All the above-mentioned video file types support the H264 video encoding format. mkv video file type supports only in H265 video encoding format.

Note: To view the recorded videos in Jetson AGX Orin™ development kit, please ensure the GStreamer plugins are properly installed. Please refer to the *e-CAM80_CUOAGX_GStreamer_Usage_Guide.pdf* to download the required plugins.

Multi Session

The customized eCAM_argus_camera application provides multi session feature to run multiple cameras simultaneously. You can enable the multi session mode from the Capture drop-down list box provided in the GTK.

When you click the **Capture** drop-down list box provided in the GTK, the available modes will appear as shown below.

Figure 21: Multi Session Modes

Multicamera Synchronization

This IMX415 camera module can be configured either as master or slave to stream the cameras in a synchronized manner. The camera module provides synchronization in which one camera acts as a master and the other cameras act as slave driven by the master's signal. By default, the application is set to stream in master mode. You can configure any one camera as master and others as slave.

Note: Only one camera should act as master at the time of multicamera synchronization and the other connected cameras should act as slave. For example, if four cameras are connected, configure first camera as master and must configure other three cameras as slave.

1. Run the following command, to configure camera as either master or slave.

```
v4l2-ctl -d <n> -c master_slave_mode=<0/1>
```

here,

<n> - refers to the device node.

<0/1> - ‘0’ to configure as master and ‘1’ to configure as slave.

2. Run the following command, to verify the value of the master-slave select control,

```
v4l2-ctl -d <n> -l | grep "master_slave_mode"
```

The output message appears as shown below.

```
master_slave_mode 0x009a2015 (intmenu): min=0 max=1
default=0 value=1
```

Note: By default, All the devices are configured as master.

3. Run the following command to launch the eCAM_argus_camera application.

Note: First, you must stream on the Master camera and then slave cameras.

```
eCAM_argus_camera --device=<n>
```

Where **<n>** refers to the master camera device node.

After master camera streamed ON, you can use Multi Session option.

© Copyright e-con Systems. 2024. All rights reserved.

Comparing eCAM_Argus_Camera with argus_camera by NVIDIA

This section describes the comparison of eCAM_Argus_Camera with argus_camera.

Users familiar with NVIDIA® Jetson AGX Orin™ Development Kit may be familiar with NVIDIA® default camera application, argus_camera.

Additional features in NVIDIA® argus_camera and not in e-con Systems eCAM_Argus_Camera are as follows:

- AC region horizontal
- AC region vertical
- YUV Format
- Focus Position
- Aperture position
- Aperture Fnum
- Aperture motor speed

Troubleshooting

In this section, you can view the list of commonly occurring issues and their troubleshooting steps.

When I capture JPEG image and leave the application on free running for one or two hours, I am getting a JPEG encoder timeout?

This is known limitation. Once the JPEG encoder timeout occurs, the streaming will be fine. But when you try to capture an image, the application may hang.

To recover the application, you must kill the application and run the following command.

```
$ sudo service nvargus-daemon restart
```

© Copyright e-con Systems. 2024. All rights reserved.

FAQ

1. **How can I get the updated package?**

   To download the updated package, please refer to the "Download Release Package" section in the *e-CAM80_CUOAGX_Release_Notes_<Rev>.pdf*

What's Next?

After understanding the usage of eCAM_Argus_Camera application, you can refer to the following documents to understand more about e-CAM80_CUOAGX.

- *e-CAM80_CUOAGX Release Package Manifest*
- *e-CAM80_CUOAGX Datasheet*

© Copyright e-con Systems. 2024. All rights reserved.

Glossary

GIMP: GNU Image Manipulation Program.

GNU: GNU's Not Unix.

GTK: GIMP Toolkit.

ISP: Image Signal Processor.

MIPI: Mobile Industry Processor Interface.

Support

Contact Us

If you need any support on e-CAM80_CUOAGX product, please contact us using the Live Chat option available on our website - https://www.e-consystems.com/

Creating a Ticket

If you need to create a ticket for any type of issue, please visit the ticketing page on our website - https://www.e-consystems.com/create-ticket.asp

RMA

To know about our Return Material Authorization (RMA) policy, please visit the RMA Policy page on our website - https://www.e-consystems.com/RMA-Policy.asp

General Product Warranty Terms

To know about our General Product Warranty Terms, please visit the General Warranty Terms page on our website - https://www.e-consystems.com/warranty.asp

© Copyright e-con Systems. 2024. All rights reserved.

Revision History

Rev	Date	Description	Author
1.0	10-Mar-2023	Initial Draft	Product Marketing Team
1.1	18-Jan-2024	Updated to L4T35.4.1 for Jetson AGX Orin™/ Xavier™ Development Kit, Added “Multi Exposure, Multi Session and Multicamera Synchronization” sections and Updated Images	Camera Dev Team
1.2	09-Apr-2024	Updated Images & removed “Exposure compensation and multi-exposure” sections	Camera Dev Team
1.3	18-Jun-2024	Updated to L4T36.3.0	Camera Dev Team